

Riphah International University
Riphah Institute of Systems Engineering
MS Information Security/ Data Science/Artificial Intelligence
Machine Learning, Fall 2026

Assignment 1

Date Delivered: Monday, September 28, 2026

Return Date: Monday, October 12, 2026

Maximum Marks 60 (30+30)

The presentations will start on 17 October.

Important Note:

While completing this assignment, you may encounter concepts, methods or Python functions that have not yet been discussed in class. This is intentional. At the MS level, you are expected to explore unfamiliar concepts independently, consult credible resources and apply what you learn appropriately. Use this assignment as an opportunity to demonstrate your problem-solving, research and technical skills. Clearly cite the resources consulted and disclose any use of generative AI.

Q.1: Multiple Linear Regression

Select a dataset from a reliable open-source platform that represents a regression problem. The dataset must contain one continuous target variable, at least three numerical predictor variables and preferably at least 200 observations.

Proceed as follows:

1. State the regression problem clearly. Identify the target variable, predictors and intended unit of prediction.
2. Describe the dataset, including:
 - Source and URL
 - Number of observations
 - Number of variables;
 - Description and units of each selected variable.

3. Perform an initial examination of:
 - Data types (Numerical/ Categorical)
 - Missing values (How many and which predictor variables)
 - Duplicate records (How many)
 - Potential outliers (How many)
4. Split the dataset into training and test sets. Develop an appropriate preprocessing procedure for missing values, outliers and feature scaling.
5. Examine multicollinearity among the predictors using.
 - a correlation matrix; and/or
 - VIF, condition number or singular values.
6. Develop a multiple linear regression model using at least two of the following computational approaches:
 - Normal-equation solution
 - QR decomposition
 - Singular Value Decomposition
 - Batch gradient descent
 - Stochastic gradient descent
7. If GD or SGD is selected:
 - Standardize the predictors
 - Report the learning rate
 - State the stopping criterion
 - Report the number of iterations or epochs
8. Evaluate each implementation on the training and test sets using:
 - MAE
 - RMSE
 - R^2 /Adjusted R^2
9. Present appropriate visualizations, including:
 - Actual versus predicted values
 - Residuals versus predicted values
 - Residual distribution
10. Compare the selected computational approaches in terms of:
 - Estimated coefficients
 - Predictions and evaluation metrics
 - Sensitivity to scaling/normalization.

Deliverables

1. A concise technical report covering all required steps, results and interpretations and self-reflection
2. A properly documented and reproducible Python program.
3. A short declaration describing any use of generative AI.
4. Presentation

Q.2: Polynomial Regression with Multiple Predictors

You may use the same dataset, training–test split and preprocessing principles employed in Q.1.

Proceed as follows:

1. Generate polynomial features for degrees: 2, 3, 4 or higher if required.
3. Report:
 - Number of predictors;
 - Number of polynomial features generated at each degree;
 - Examples of individual-power terms;
 - Examples of cross or interaction terms.
4. Develop a polynomial regression model for each degree.
5. Select the polynomial degree using k -fold cross-validation on the training data. The test data must not be used to select the degree.
6. For the selected model, report:
 - Intercept
 - Coefficients of all individual-power terms;
 - Coefficients of all interaction terms;
 - Training and test MAE
 - Training and test RMSE
 - Training and test R^2
7. Present appropriate visualizations, including:
 - Training and validation error against polynomial degree
 - Actual versus predicted values
 - Residuals versus predicted values

8. Compare the selected polynomial model with the multiple linear regression model from Q.1.
9. Discuss:
 - Whether polynomial regression improves generalization
 - Evidence of underfitting or overfitting
 - Growth in the number of generated features
 - Multicollinearity among polynomial terms

Deliverables

1. A concise technical report covering methodology, results and interpretation.
2. A properly documented and reproducible Python program.
3. A short declaration describing any use of generative AI.
4. Presentation